

AUTONOMOUS SMART SPRAYING ROVER FOR CROPS

A PROJECT REPORT

Submitted to Government College of engineering, Jalgaon in Partial Fulfilment of
the Requirements for the Degree of BACHELOR OF TECHNOLOGY in
Instrumentation Engineering.

by

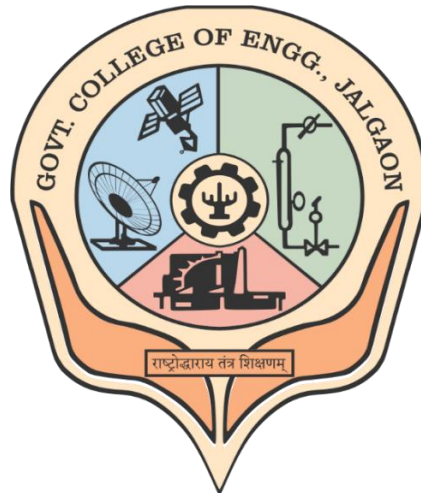
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GOVERNMENT COLLEGE OF ENGINEERING, JALGAON 425001**

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GOVERNMENT COLLEGE OF ENGINEERING, JALGAON 425001

(An Autonomous Institute of Govt. Of Maharashtra)

DEPARTMENT OF INSTRUMENTATION ENGINEERING

CERTIFICATE



This is to certify that the project entitled, "**AUTONOMOUS SMART SPRAYING ROVER FOR CROPS**" which is being submitted herewith for the award of B-TECH, is the result of the work completed by **TANUJA PATIL (2322225)**, **KUNAL PRAVIN PATIL (2322223)**, **SUMIT SANJAY DESALE (2322206)** under my supervision and guidance within the four walls of the institute and same has not been submitted elsewhere for the award of any Degree.

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DECLARATION

We hereby declare that the project entitled, “**AUTONOMOUS SMART SPRAYING ROVER FOR CROPS**” was carry out and written by us under the guidance of **Prof. P. J. Turare, Assistant professor, Department Instrumentation Engineering**, Government college of Engineering, Jalgaon. This work has not been previously formed the basis for the award of any degree or diploma or certificate nor has been submitted elsewhere for the award of any degree or diploma.

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I extend my appreciation to my teammates for their cooperation, teamwork, and dedication in completing this seminar project successfully.

Finally, I express my heartfelt gratitude to my parents and family members for their continuous encouragement and moral support. Students can add extra lines as per their need.

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ABSTRACT

This project presents a "Smart Disease-Detecting and Spraying Rover," an autonomous agricultural system designed to address the challenges of inefficient, labor-intensive disease monitoring and excessive pesticide use in modern farming.

The rover autonomously navigates through crop fields, using an onboard camera to capture real-time images of plants. These images are processed by a pre-trained machine learning model, hosted on ESP32, to detect visible symptoms of diseases. When a diseased plant is identified, the rover activates a precision spraying mechanism that targets only the affected area, which reduces pesticide usage and minimizes environmental impact.

The system's core components include an ESP32 as the central processor, a camera for vision, a motor driver (like the L293D) for mobility, and a pesticide pump for targeted application. The effectiveness of the disease detection model was demonstrated via a web-based simulation, which correctly classified grape leaves as "Healthy" or "Diseased" with high confidence.

By automating these critical tasks, the project aims to enhance productivity, reduce manual labour, and promote more sustainable farming practices.

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CHAPTER 1

INTRODUCTION

Modern agriculture faces a myriad of challenges, including the timely detection of plant diseases and the excessive use of pesticides, which can negatively impact crop yields, soil health, and the environment. Traditional methods of disease monitoring and pesticide application are often labour-intensive, imprecise, and inefficient. In response to these challenges, this project presents the Smart Disease-Detecting and Spraying Rover — an innovative, autonomous agricultural system designed to transform how disease detection and pesticide application are carried out in the field.

This smart rover integrates computer vision and machine learning technologies to detect visible symptoms of plant diseases in real time. Equipped with a camera module and a pre-trained image processing model, the system continuously scans crops as it navigates the field. When a diseased plant is identified, the rover activates a precision spraying mechanism that targets only the affected area, thereby reducing pesticide usage and minimizing environmental impact.

The mobility and control of the rover are managed using the ESP32, a low-cost and energy-efficient microcontroller with built-in Wi-Fi and Bluetooth capabilities. It is interfaced with motor drivers to ensure smooth movement across the field. Additional sensors — including infrared (IR), ultrasonic, and GPS modules — enable intelligent navigation and obstacle avoidance, ensuring safe and efficient operation under varying field conditions.

Despite its compact architecture compared to high-end processors, the ESP32 is capable of handling real-time control tasks and wireless communication efficiently. For computationally intensive tasks such as image processing, the system can either utilize optimized embedded models or offload processing to cloud or edge platforms via IoT connectivity.

By automating disease detection and pesticide application, this project aims to assist farmers in enhancing productivity, reducing manual labor, and promoting sustainable farming practices. Furthermore, the integration of IoT components enables remote monitoring, real-time data transmission, and logging, allowing farmers to make informed decisions regarding crop management.

In essence, the Smart Disease-Detecting and Spraying Rover represents a significant step toward the digitization and automation of agricultural practices, aligning with the global push for smarter and more sustainable food production system.

1.1.1 BACKGROUND

Agriculture is one of the most important sectors for food production and economic growth. Farmers often face challenges such as pest attacks, plant diseases, and weed growth, which reduce crop yield and quality. Traditionally, pesticides and fertilizers are sprayed manually using handheld pumps or tractor-mounted sprayers. These traditional methods require more labor, consume more chemicals, and may expose farmers to harmful substances.

With the advancement of robotics, automation, sensors, and Internet of Things (IoT), smart agricultural systems are becoming popular. A Smart Spraying Rover is an automated ground vehicle designed to move through crop fields and spray pesticides or fertilizers precisely where needed. It helps improve farming efficiency, reduces waste, and supports modern precision agriculture practices.

1.2 MOTIVATION

The idea of a Smart Spraying Rover is motivated by the need to solve common agricultural problems such as labour shortages, increasing farming costs, and overuse of chemicals. Manual spraying is time-consuming and physically tiring, especially in large farms. Farmers are also exposed to toxic chemicals during spraying, which can affect health.

Another motivation is environmental protection. Excessive use of pesticides can contaminate soil, water, and crops. By using smart spraying technology, chemicals can be applied only to affected areas in the required quantity. This reduces pollution and increases crop safety.

The rover also supports smart farming by reducing human effort and improving productivity with automation.

1.3 PROBLEM STATEMENT

Traditional crop spraying methods are inefficient, labour-intensive, and unsafe. Farmers often spray pesticides uniformly across the field, even in areas where treatment is not required. This leads to:

- Wastage of pesticides and fertilizers
- Increased farming cost
- Health risks to farmers due to chemical exposure
- Uneven spraying and poor crop protection
- Environmental pollution
- Difficulty in spraying large or difficult terrains

Therefore, there is a need for an automated and intelligent system that can move in crop fields and perform accurate spraying with minimal human intervention.

1.4 OBJECTIVES OF THE PROJECT

The main objectives of the Smart Spraying Rover project are:

- To design and develop an automated rover for crop spraying.
- To reduce manual labour in agricultural spraying operations.
- To provide precise and controlled spraying of pesticides/fertilizers.
- To minimize chemical wastage and reduce operational cost.
- To protect farmers from direct exposure to harmful chemicals.
- To improve crop health and productivity through timely spraying.
- To integrate sensors, controllers, and smart navigation systems.
- To promote modern precision agriculture techniques.

1.5 SCOPE OF THE PROJECT

The Smart Spraying Rover has wide scope in present and future agriculture systems. It can be used in:

- Small, medium, and large farms for spraying pesticides and fertilizers.
- Greenhouses and playhouses for controlled crop management.
- Orchards and plantations for targeted spraying.
- Weed control applications using selective herbicide spraying.
- Remote monitoring and operation using mobile app or IoT system.
- Integration with AI-based disease detection in future upgrades.
- Autonomous navigation using GPS and obstacle detection sensors.
- Data collection for soil, crop health, and field conditions.

CHAPTER 2

LITERATURE REVIEW

1.Pesticides Spraying Using Non-GPS-Based Autonomous Drone Kaushik Baidya Dept. of ESE, National Institute of Electronics and Information Technology (NIELIT) Aurangabad, Maharashtra, India

For a country like India where the land size is small and there is no accurate GPS mapping available, the designed UAV can map the field accurately and follow the designated path for pesticide spraying. UAVs with artificial intelligence capabilities. The methodology proposed in this research is novel in comparison to existing unmanned aerial vehicles utilized in agriculture. Additional sensors can make the entire process of field monitoring considerably simpler and easier.

2.2024 International Conference on Advanced Information Scientific Development (ICAISD)Performance Analysis of MobileNET on Grape Leaf Disease Detection 1st Moehammad Sarosa,Department of Electrical Engineering,Malang State Polytechnic Malang, Indonesia

The early identification of grapevine diseases is an important step in maintaining plant health and increasing yields.Grape vines are agricultural commodities with high economic value but are susceptible to various types of diseases. Disease detection technology using convolutional neural networks (CNN),one of which uses the MobilNet architecture, shows good results.This research aims to demonstrate the performance analysis ofMobilNet in detecting diseases in grape leaves. The training process uses a dataset of 1600 images of leaves infected with three diseases: Black-Rot, Esca, leaf blight, and one healthy leaf. MobilNet was selected because of its high ability to classify images with limited computational resources. Thus, this method can help farmers implement preventive measures in a timely manner. MobilNet proved to be efficient and effective in detecting grape leaf diseases, achieving an average validation accuracy of 98% and a testing accuracy of 97%. This model can respond quickly to improve the quality and quantity of the grape harvest and prevent the spread of diseases in grape plants to minimize economic losses.

3. Early Detection of Grapes Diseases UsingMachine Learning and IoT Suyash S. Patil1, Sandeep A. Thorat21, 2 Department of Computer Science EngineeringRajarambapu Institute of Technology, RajaramnagarIslampur, Sangli (MS) 415409 India

cultivation has social and economic importance in India. In India, Maharashtra ranks first in grapes production. Over the last few years the quality of grapes has degraded because of many reasons. One of the important causes is diseases on grapes. To prevent diseases farmers spray huge amount of pesticides, which result in increasing the cost of production. Also farmers are unable to identify the diseases manually. The diseases are identified only after the infection, but it takes up a lot of time and have adverse effects on vineyard. The proposed work is to develop a monitoring system which will identify the chances of grape diseases in its early stages by using Hidden Markov Model provides alerts via SMS to the farmer and the expert. The system includes temperature, relative humidity, moisture, leaf wetness sensor.

4. Design and Investigation of Automatic Pest Detection and Pesticide Spraying Device **S.Ramanathan, Mechanical Engineering Sri Sairam Engineering College, Chennai, India**

Pest detection in paddy fields is a significant challenge in agriculture, so effective measures to combat the infestation while minimising pesticide use should be developed. Image analysis techniques are widely used in agricultural science, providing maximum crop protection and ultimately leading to better crop management and production. Pest infestations are currently monitored by hand, but automated monitoring is being developed to reduce human effort and errors. It is possible to accomplish this by using a camera to capture input images and then analyzing them with machine learning. A motor driver is used to build an agricultural robot, and a Raspberry Pi3 is used as the processor or embedded system. For machine learning, we use Python code to train the robot with predefined images. Pest infestations are currently monitored by hand, but automated monitoring is being developed to reduce human effort and errors

5. Drone Technology Enabled Leaf Disease Detection and Analysis system for Agriculture Applications **Lakshmi Narayana Thalluri, Sai Divya Adapa, Priyanka D, Sri Nithya N, Yasmeen, Addepalli V S Y Narayana Sarma, Srihakolanu Naga Venkat Identification of plant**

diseases plays a crucial role in preventing the loss of the yield and quantity of agriculture. A visually observable pattern on leaves helps to study the diseases in plants. Agricultural drone are highly efficient and their usage has been expanded to many areas in agriculture including fertilizer spraying, sowing seeds, growth monitoring, and mapping.

CHAPTER 3

THEORETICAL BACKGROUND

The Autonomous Smart Spraying Rover for Precision Pesticide Application is based on the concept of precision agriculture, robotics, automation, image processing, and embedded systems. The main purpose of this system is to reduce pesticide wastage, improve spraying accuracy, and minimize human effort in agricultural operations.

Traditional pesticide spraying methods are generally manual and require significant labor. In many cases, pesticides are sprayed uniformly across the entire field, even when only a small portion of the crop is affected by disease or pests. This leads to excessive chemical usage, increased farming cost, environmental pollution, and health risks to farmers. To overcome these limitations, smart agricultural systems using automation and artificial intelligence are becoming increasingly important.

The smart spraying rover works as an autonomous ground vehicle that can move through crop fields and spray pesticides only where required. The rover uses sensors, a camera module, microcontroller, motor driver, and spraying mechanism to perform intelligent spraying operations. The complete system combines hardware and software technologies for efficient field operation.

The system mainly operates on the principle of disease detection and precision spraying. A camera mounted on the rover continuously captures images of crop leaves while the rover moves in the field. These images are processed using machine learning and image processing techniques. A Convolutional Neural Network (CNN) model is commonly used for disease identification because of its high accuracy in image classification tasks. The trained model can identify healthy and diseased leaves based on visual symptoms such as spots, discoloration, fungal infection, and leaf damage.

After detecting the disease, the controller sends a signal to activate the pesticide spraying unit. The spraying pump sprays pesticide only on the infected plant area, which helps reduce chemical wastage and improves crop protection. This method is known as targeted or precision spraying.

The rover movement is controlled using DC geared motors connected through a motor driver such as L293D or L298N. The motor driver acts as an interface between the microcontroller and motors by providing the required current and voltage for motor operation. The rover can move forward, backward, left, and right according to programmed instructions.

The ESP32 microcontroller plays a major role in controlling the complete operation of the rover. It processes sensor signals, controls motor movement, activates the relay and pump, and manages wireless communication. ESP32 is widely used because of its low power consumption, compact size, and built-in Wi-Fi and Bluetooth connectivity.

Obstacle detection is achieved using ultrasonic sensors. The ultrasonic sensor works on the SONAR principle. It transmits ultrasonic waves and receives reflected waves from nearby obstacles. By measuring the time difference between transmission and reception, the sensor calculates the distance to the obstacle. This helps the rover avoid collisions during field navigation.

The spraying mechanism consists of a pesticide tank, diaphragm pump, relay module, and nozzle. The relay acts as an electrically operated switch that controls the ON/OFF operation of the pump. When the controller detects a diseased plant, the relay activates the pump and pesticide is sprayed through the nozzle in a controlled manner.

Wireless communication and IoT technology can also be integrated into the system for remote monitoring and control. Farmers can monitor rover operation, spraying status, and disease detection information using a mobile application or cloud platform. This supports modern smart farming and precision agriculture practices.

The Autonomous Smart Spraying Rover provides several benefits such as:

- Reduction in pesticide wastage
- Lower farming cost
- Reduced human exposure to chemicals
- Uniform and accurate spraying
- Improved crop health and productivity
- Support for sustainable agriculture

CHAPTER 4

COMPONENTS AND THEIR DESCRIPTION

4.1 PROJECT SPECIFICATION:

The Smart Disease-Detecting and Spraying Rover is designed as an autonomous system that integrates hardware and software components for real-time plant disease detection and targeted pesticide application. The specifications are categorized into functional and non-functional aspects to clearly define the system requirements and expected performance.

4.1.1 FUNCTIONAL SPECIFICATIONS

- **Real-Time Disease Detection**

- Captures plant images using an onboard camera.
- Processes images using machine learning and image processing algorithms.
- It identifies visible symptoms of plant diseases with a pre-trained classification model.

- **Autonomous Navigation**

- Moves across agricultural fields without manual intervention.
- Utilizes motor drivers for mobility and control.
- Incorporates sensors for obstacle detection and avoidance.

- **Selective Pesticide Spraying**

- Activates the spraying unit only when a diseased plant is detected.
- Applies pesticides precisely to the infected area to reduce wastage.

- **Data Monitoring and Logging**

- Logs disease detection events with time and (optionally) GPS location.

4.2 BLOCK DIAGRAM:

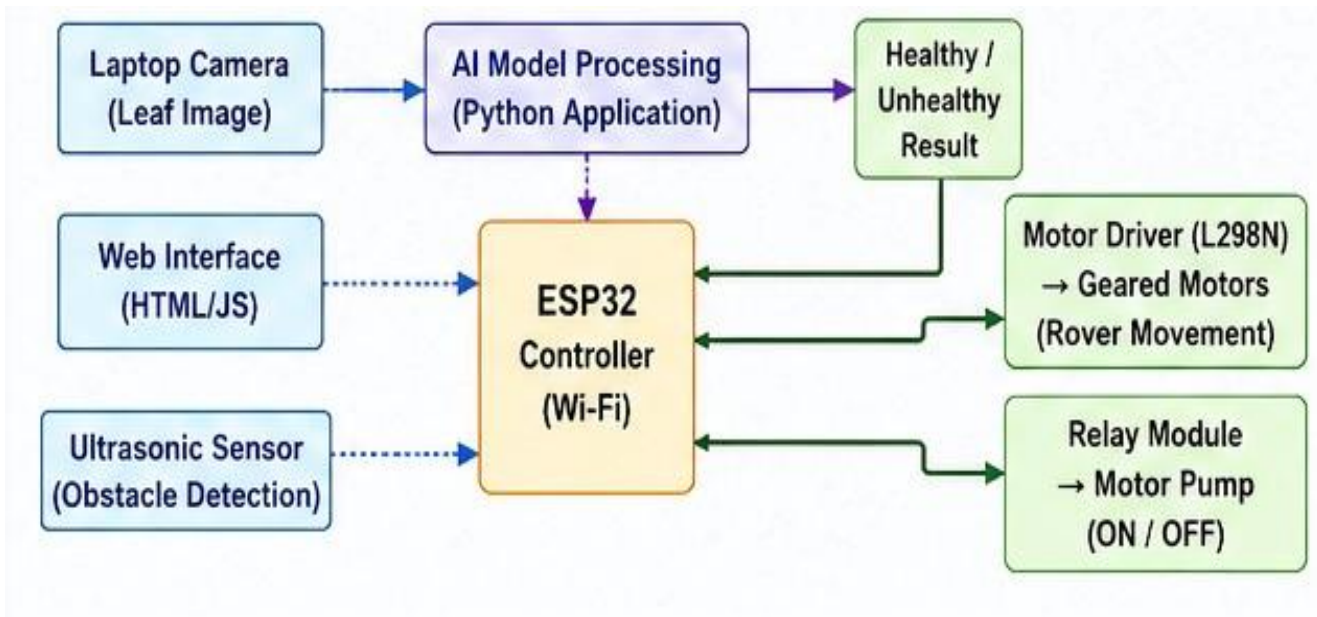


Fig: 4.2.1 Block Diagram of Model

4.3 DESCRIPTION OF BLOCKS:

The system block diagram illustrates the functional architecture of the Smart Disease- Detecting and Spraying Rover. It highlights the key components and their interactions in performing real-time plant disease detection and targeted pesticide application. The entire process is powered and controlled through a coordinated set of modules, as described below:

4.3.1 Power supply:

The power supply serves as the primary energy source for the rover and is responsible for providing electrical power to all other components. Typically, this is achieved using a rechargeable Lithium-ion battery pack, chosen for its high energy density, long life cycle, and portability. The power system is carefully designed to deliver consistent voltage and current levels to sensitive electronics like the microcontroller, as well as power-hungry components such as the motors and sprayer. Voltage regulation circuitry is often included to ensure that power delivery is stable and within safe operating limits, protecting components from damage due to surges or drops. This ensures uninterrupted operation in field conditions where power availability is limited.

4.3.2 Microcontroller ESP 32:

The ESP32-WROOM is a highly integrated, low-power System-on-Chip (SoC) module developed by Espressif Systems, designed for a wide range of Internet of Things (IoT) applications. At its core, it features the Xtensa® Dual-Core 32-bit LX6 microprocessor, capable of operating at frequencies up to 240 MHz, which provides a significant performance boost over its predecessor, the ESP8266. One of its most defining features is the integration of dual-mode wireless connectivity, supporting both 2.4 GHz Wi-Fi and Bluetooth (Classic and Low Energy). This allows the module to act as a standalone station or a bridge between legacy Bluetooth devices and modern cloud-based networks.



Fig: 4.3.2 Microcontroller ESP 32

4.3.3 Motor Driver:

The motor driver is an electronic circuit that controls the motion of the rover by regulating the power delivered to the motors. It acts as an intermediary between the microcontroller and the DC motors, as the microcontroller alone cannot supply the high current needed for motor operation. The motor driver receives signals from the microcontroller regarding speed and direction, and uses these to drive the motors accordingly. Components like the L298N or L293D are commonly used motor drivers in robotics. With its ability to manage forward and reverse movements and adjust motor speed via PWM (Pulse Width Modulation), the motor driver enables the rover to navigate autonomously through crop fields.



Fig: 4.3.3 Motor Driver

4.3.4 Camera (Disease Detection):

The camera serves as the rover's vision system, continuously capturing images of the plants as it moves along its path. These images are vital for the disease detection process. Positioned strategically to ensure maximum visibility of leaves and stems, the camera captures high-resolution images suitable for detailed analysis. A standard USB webcam or a dedicated Pi Camera module can be used, depending on the processing board. The camera must function reliably in outdoor lighting conditions and provide consistent image quality to support accurate image processing and machine learning analysis. It plays a crucial role in enabling the rover to "see" and understand its environment.



Fig: 4.3.4 ESP32 Camera Module

4.3.5 Image Processing and Machine Learning Model:

This component is the core of the rover's intelligence, where the real-time analysis of plant images takes place. Using image processing techniques and a pre-trained machine learning model, this system analyzes the visual data collected by the camera to detect symptoms of plant diseases. The model, often based on convolutional neural networks (CNNs), can identify issues like leaf spots, discoloration, fungal growth, and wilting. Once a diseased plant is detected, a signal is sent to

initiate the spraying mechanism. This module is typically implemented on a Raspberry Pi using frameworks like OpenCV, Tensor Flow, or PyTorch, and ensures that only affected plants are targeted for treatment.

4.3.6 Spraying Mechanism (Pesticide Nozzle):

The spraying mechanism is responsible for the physical application of pesticide to diseased plants. It consists of a pesticide reservoir, a pump (such as a peristaltic or diaphragm pump), and a nozzle or sprayer. Controlled by the microcontroller, this mechanism activates only when a plant is confirmed to be infected, ensuring minimal chemical usage. The system is designed to provide precise, localized spraying to reduce environmental harm and cost. A solenoid valve may be used to control the flow of liquid, and the nozzle can be configured to deliver a fine mist or a directed spray, depending on the crop type and application method.



Fig: 4.3.6 Spraying Mechanism (Pesticide Nozzle)

4.3.7 Detected Disease = Trigger Spray:

This is the logical control point in the system where the decision to apply pesticide is executed.

Once the machine learning model confirms the presence of disease in the plant, the system Generates a trigger signal that activates the spraying mechanism. This ensures a closed feedback loop between disease detection and pesticide application. By tying the detection directly to actuation, the rover ensures that pesticide is applied immediately and only where it is needed. This block embodies the automation aspect of the rover, allowing it to operate independently across large agricultural areas while maintaining precision.

4.3.8 Pump:

The sprayer unit used in this project is a 12V DC Mini Diaphragm Pump, commonly utilized in agricultural spraying, fluid transfer, and portable misting applications. It is a compact, lightweight, and efficient pump designed for continuous and automated pesticide spraying.

This pump operates on the diaphragm principle, where a reciprocating diaphragm creates pressure differences to draw liquid from the reservoir and discharge it at high pressure. It offers a flow rate of approximately 6–8 L/min and an operating pressure in the range of 0.4– 0.6 MPa, making it suitable for precision spraying in agricultural fields. The pump includes an automatic pressure switch, which stops the motor when the outlet is closed or pressure builds up beyond safe limits. This ensures energy saving, prevents motor overload, and enhances overall safety. Being self-priming, the pump can lift water even when mounted above the tank, improving installation flexibility on the rover.



Fig: 4.3.8 Pump

4.3.9 Battery:

The project uses a 12V, 8Ah sealed lead-acid (SLA) rechargeable battery, which serves as the primary power source for the rover. This type of battery is widely used in portable electronic systems due to its reliability, stable output, and maintenance-free operation. The sealed lead-acid design ensures safe usage, prevents leakage, and allows the battery to operate in various orientations. With an 8Ah capacity, the battery can supply sufficient current to drive the DC motors, microcontroller unit, sensors, and the spraying pump for extended periods.



Fig: 4.3.9 Battery

4.3.10 Johnson Gear Motor:

It is a simple DC motor featuring metal gearbox for driving the shaft of the motor, so it is a mechanically commutated electric motor which is powered from DC supply. The Johnson Geared Motors are known for their compact size and massive torque-speed characteristic.

The Johnson Motor comes with side shaft also known as an off-centered shaft and six M3 mounting holes. The shaft of the motor equips metal bushes which makes these DC gear motors Shaft wear resistant. The shaft of the motor has a hole for better coupling.

The motor will run smoothly between the voltage range 6 to 18 V DC and give you 300 RPM at 12V supply. It provides the torque of 2.2 Kg-cm at 300 RPM. The Johnson Geared Motor gives very good torque at an affordable price hence they are widely applicable in Pan/Tilt camera, auto shutter, welding machines, water meter IC card, grill, oven, cleaning machine garbage disposers, household appliances, slot machines, money detector, automatic actuator, coffee machine, towel disposal, lighting coin refund devices, the peristaltic pump and so on.



Fig: 3.3.7 Johnson Gear Motor

4.3.11 Robot wheels:

This is a lightweight plastic and rubber robot wheel commonly used for small rovers and robotic vehicles. It features a durable plastic hub with a rubber outer tread that provides good traction on uneven surfaces. The wheel is designed to fit standard DC geared motors, making it suitable for smooth movement and stable navigation in robotics applications.



Fig: 4.3.11 Robotic Wheels

4.3.12 Ultrasonic sensor:

This HC-SR04-Ultrasonic Range Finder is a very popular sensor that is found in many applications where it requires measuring distance and detecting objects. The module has two eyes like projects in the front which forms the Ultrasonic transmitter and Receiver. The HC-SR04 ultrasonic sensor uses sonar to determine the distance to an object like bats or dolphins do.

This Ultrasonic Sensor module is a transmitter, a receiver, and a control circuit in one single pack!! It has very handy and compact construction. It offers excellent range accuracy and stable readings in an easy-to-use package. Its operation is not affected by sunlight or black material like Sharp rangefinders are (although acoustically soft materials like cloth can be difficult to detect).

The Trigger and the Echo pins are the I/O pins of this module and hence they can be connected to the I/O pins of the microcontroller/Arduino. When the receiver detects the return wave the Echo pin goes high for a particular amount of time which will be equal to the time taken for the wave to return back to the sensor.



Fig: 4.3.12 Ultrasonic Sensor

4.3.13 Relay:

The component shown is a 5V Single-Channel Relay Module, commonly used in embedded systems for switching high-power devices using low-power control signals from microcontrollers such as Arduino, ESP32, or Raspberry Pi. The module features an SRD-05VDC-SL-C electromagnetic relay, capable of switching loads up to 10A at 250VAC or 10A at 30VDC, making it suitable for controlling motors, pumps, solenoids, and other high-current devices in the project.

The relay module includes a driver circuit, consisting of a transistor, diode, and indicator LED. This circuitry allows the relay coil to be activated using a small 3.3V or 5V digital signal while providing electrical isolation between the control side and the load side. The built-in flyback diode protects the microcontroller from voltage spikes generated during relay switching, improving the reliability of the system.

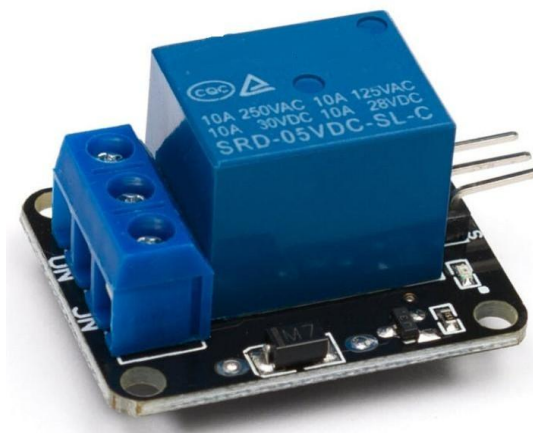


Fig: 4.3.13 Relay

CHAPTER 5

SYSTEM WORKING

5.1 WORKING PRINCIPLE:

The Autonomous Smart Spraying Rover for Precision Pesticide Application works on the principle of automated disease detection, intelligent navigation, and precision pesticide spraying. The system combines image processing, embedded control, sensors, and spraying mechanisms to perform efficient agricultural operations with minimum human intervention.

The operation of the rover begins when power is supplied through the rechargeable battery. The battery provides electrical energy to all components including the ESP32 microcontroller, motor driver, sensors, camera module, and spraying pump. The ESP32 acts as the main controller of the entire system and manages all operations of the rover.

Initially, the rover starts moving through the crop field using DC geared motors connected through the motor driver module. The motor driver controls the direction and speed of the rover according to the control signals received from the ESP32. The rover is designed to move autonomously between crop rows for continuous field monitoring.

As the rover moves, the camera module mounted on the rover captures real-time images of crop leaves and plants. These images are processed using image processing and machine learning techniques. A pre-trained Convolutional Neural Network (CNN) model analyzes the captured images and identifies whether the plant is healthy or affected by disease.

The disease detection process is based on visual symptoms such as:

- Leaf spots
- Color changes
- Fungal infections
- Dry or damaged leaf regions

When the CNN model detects a diseased plant, the processed information is sent to the ESP32 controller. The controller then makes a decision regarding pesticide spraying.

After confirmation of disease detection, the ESP32 activates the relay module connected to the pesticide pump. The relay acts as an electronic switch that turns the pump ON and OFF automatically. The pump draws pesticide solution from the storage tank and sprays it through the nozzle directly onto the affected plant area. This targeted spraying mechanism reduces unnecessary pesticide usage and minimizes environmental pollution.

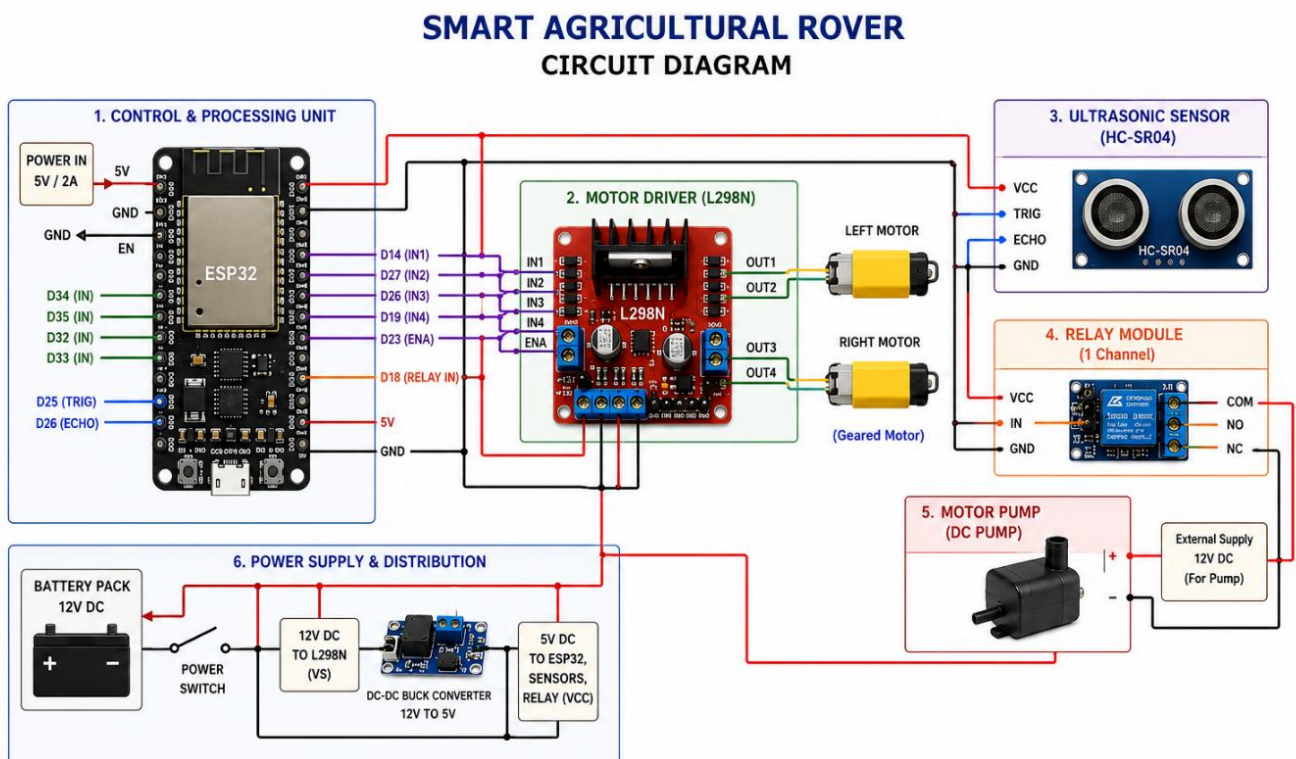
The rover also includes an ultrasonic sensor for obstacle detection and safe navigation. The ultrasonic sensor continuously measures the distance between the rover and nearby objects using ultrasonic waves. If any obstacle is detected within a specified range, the ESP32 stops or redirects the rover to avoid collision.

The complete operation follows the sequence below:

1. Power ON the system
2. Rover starts moving through the field
3. Camera captures crop images
4. CNN model analyzes images
5. Disease detection decision is made
6. Relay activates pesticide pump
7. Precision spraying is performed
8. Ultrasonic sensor checks for obstacles
9. Rover continues navigation and monitoring

The system may also support wireless communication using Wi-Fi or Bluetooth for remote monitoring and control through a mobile application or IoT platform. Farmers can observe rover status, disease detection results, and spraying operations in real time.

5.2 CIRCUIT DIAGRAM



Fig; 5.2 Circuit Diagram

7.3 PROGRAM OF MODEL

- PROGRAM FOR IMAGE DETECTION

```
import cv2
import requests
import numpy as np

ESP32_IP = "192.168.4.1"

cap = cv2.VideoCapture(0, cv2.CAP_DSHOW)

cap.set(3, 320)
cap.set(4, 240)

cap.set(cv2.CAP_PROP_BUFFERSIZE, 1)

while True:

    ret, frame = cap.read()

    if not ret:
        break

    # Resize image
    img = cv2.resize(frame, (128,128))

    # Convert to grayscale
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Calculate brightness
    brightness = np.mean(gray)

    # SIMPLE AI LOGIC
    # Dark leaf = unhealthy
    # Bright leaf = healthy

    if brightness < 100:

        result = "UNHEALTHY"

    try:
        requests.get(
```

```

        f"http://{ESP32_IP}?cmd=unhealthy"
    )
except:
    pass

else:

    result = "HEALTHY"

    try:
        requests.get(
            f"http://{ESP32_IP}?cmd=healthy"
        )
    except:
        pass

cv2.putText(
    frame,
    result,
    (20,50),
    cv2.FONT_HERSHEY_SIMPLEX,
    1,
    (0,255,0),
    2
)

cv2.imshow("Leaf Detection", frame)

if cv2.waitKey(1) == ord('q'):
    break

cap.release()

cv2.destroyAllWindows()

```

- **PROGRAM FOR CONTROL**

```

import network
import socket
from machine import Pin, time_pulse_us
from time import sleep, sleep_us

```

```

# =====
# FRONT MOTOR DRIVER
# =====

f_in1 = Pin(23, Pin.OUT)
f_in2 = Pin(22, Pin.OUT)
f_in3 = Pin(21, Pin.OUT)
f_in4 = Pin(19, Pin.OUT)

# =====
# REAR MOTOR DRIVER
# =====

r_in1 = Pin(18, Pin.OUT)
r_in2 = Pin(5, Pin.OUT)
r_in3 = Pin(17, Pin.OUT)
r_in4 = Pin(16, Pin.OUT)

# =====
# PUMP RELAY
# =====

pump = Pin(2, Pin.OUT)

# Relay OFF initially
pump.value(1)

# =====
# ULTRASONIC SENSOR
# =====

trig = Pin(32, Pin.OUT)
echo = Pin(33, Pin.IN)

# =====
# AI RESULT VARIABLE
# =====

ai_result = "NO DETECTION"

# =====
# MOTOR FUNCTIONS

```

```
# =====
```

```
def stop():
```

```
    pins = [  
        f_in1, f_in2, f_in3, f_in4,  
        r_in1, r_in2, r_in3, r_in4  
    ]
```

```
    for pin in pins:  
        pin.value(0)
```

```
def forward():
```

```
    stop()
```

```
    f_in1.value(1)  
    f_in2.value(0)
```

```
    f_in3.value(1)  
    f_in4.value(0)
```

```
    r_in1.value(1)  
    r_in2.value(0)
```

```
    r_in3.value(1)  
    r_in4.value(0)
```

```
def backward():
```

```
    stop()
```

```
    f_in1.value(0)  
    f_in2.value(1)
```

```
    f_in3.value(0)  
    f_in4.value(1)
```

```
    r_in1.value(0)  
    r_in2.value(1)
```

```
    r_in3.value(0)
```

```

    r_in4.value(1)

def left():

    stop()

    f_in1.value(0)
    f_in2.value(1)

    f_in3.value(1)
    f_in4.value(0)

    r_in1.value(0)
    r_in2.value(1)

    r_in3.value(1)
    r_in4.value(0)

def right():

    stop()

    f_in1.value(1)
    f_in2.value(0)

    f_in3.value(0)
    f_in4.value(1)

    r_in1.value(1)
    r_in2.value(0)

    r_in3.value(0)
    r_in4.value(1)

# =====
# PUMP FUNCTIONS
# =====

def pump_on():

    pump.value(0)

```

```

    print("Pump ON")

def pump_off():

    pump.value(1)

    print("Pump OFF")

# =====
# DISTANCE FUNCTION
# =====

def get_distance():

    trig.off()
    sleep_us(2)

    trig.on()
    sleep_us(10)

    trig.off()

    try:

        duration = time_pulse_us(echo, 1, 30000)

        distance = (duration * 0.0343) / 2

        return round(distance, 2)

    except:

        return 0

# =====
# WIFI ACCESS POINT
# =====

ssid = "ESP32_CAR"
password = "12345678"

ap = network.WLAN(network.AP_IF)

```



```

ap.active(True)

ap.config(essid=ssid, password=password)

while not ap.active():
    pass

print("WiFi Name:", ssid)
print("IP Address:", ap.ifconfig()[0])

# =====
# SERVER
# =====

addr = socket.getaddrinfo('0.0.0.0', 80)[0][-1]

server = socket.socket()

server.bind(addr)

server.listen(5)

print("Server Running...")

# =====
# MAIN LOOP
# =====

while True:

    cl, addr = server.accept()

    request = cl.recv(1024).decode()

    print(request)

    global ai_result

    # =====
    # CAR CONTROL
    # =====

```

```
if 'cmd=forward' in request:
```

```
    forward()
```

```
elif 'cmd=backward' in request:
```

```
    backward()
```

```
elif 'cmd=left' in request:
```

```
    left()
```

```
elif 'cmd=right' in request:
```

```
    right()
```

```
elif 'cmd=stop' in request:
```

```
    stop()
```

```
# =====
```

```
# PUMP MANUAL CONTROL
```

```
# =====
```

```
elif 'cmd=pump_on' in request:
```

```
    pump_on()
```

```
elif 'cmd=pump_off' in request:
```

```
    pump_off()
```

```
# =====
```

```
# AI CONTROL
```

```
# =====
```

```
elif 'cmd=healthy' in request:
```

```
    ai_result = "HEALTHY"
```

```
    pump_off()
```

```
elif 'cmd=unhealthy' in request:
```

```
    ai_result = "UNHEALTHY"
```

```
    pump_on()
```

```

# =====
# DISTANCE
# =====

distance = get_distance()

# =====
# HTML WEBPAGE
# =====

html = """<!DOCTYPE html>

<html>

<head>

<title>AI AGRICULTURE ROVER</title>

<meta name="viewport" content="width=device-width, initial-scale=1">

<style>

body{
    text-align:center;
    font-family:Arial;
    background:#dff0d8;
}

button{
    width:140px;
    height:60px;
    font-size:18px;
    margin:6px;
    border:none;
    border-radius:10px;
    background:#27ae60;
    color:white;
}

.distance{
    font-size:28px;
    color:red;

```

```
    margin:20px;
}
```

```
.ai{
    font-size:35px;
    color:blue;
    margin:20px;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>AI AGRICULTURE ROVER</h1>
```

```
<div class="distance">
```

```
Distance: "" + str(distance) + "" cm
```

```
</div>
```

```
<h2>VEHICLE CONTROL</h2>
```

```
<button onclick="send('forward')">Forward</button>
```

```
<br><br>
```

```
<button onclick="send('left')">Left</button>
```

```
<button onclick="send('stop')">Stop</button>
```

```
<button onclick="send('right')">Right</button>
```

```
<br><br>
```

```
<button onclick="send('backward')">Backward</button>
```

```
<h2>PUMP CONTROL</h2>
```

```
<button onclick="send('pump_on')">Pump ON</button>
```

```
<button onclick="send('pump_off')">Pump OFF</button>
```

```
<h2>AI RESULT</h2>
```

```
<div class="ai">
```

```
"" + ai_result + ""
```

```
</div>
```

```
<script>
```

```
function send(cmd)
{
    fetch('/?cmd=' + cmd);
}
```

```
</script>
```

```
</body>
```

```
</html>
```

```
""
```

```
# =====
```

```
# SEND RESPONSE
```

```
# =====
```

```
cl.send('HTTP/1.1 200 OK\r\n')
```

```
cl.send('Content-Type: text/html\r\n')
```

```
cl.send('Connection: close\r\n\r\n')
```

```
cl.sendall(html)
```

```
cl.close()
```

CHAPTER 6

ADVANTAGES & APPLICATION

6.1 ADVANTAGES:

- **Precise Motor Control**

The L293D motor driver allows the Raspberry Pi to control both the direction and speed of DC motors using simple logic and PWM (Pulse Width Modulation). This enables smooth acceleration, deceleration, and bidirectional movement.

- **Cost-Effective and Readily Available Components**

Both the Raspberry Pi and L293D are affordable and easily sourced, making this setup perfect for students, hobbyists, and educators working on low-budget robotics and automation projects.

- **Compact and Lightweight Design**

The entire setup is small and space-efficient, ideal for compact robots, drones, and embedded systems where saving space is critical.

- **Python Programmability and Linux Support**

Raspberry Pi runs a full Linux OS (Raspberry Pi OS) and supports Python, allowing for easy coding and integration with advanced algorithms, machine learning models, and sensors.

- **Dual Motor Control Capability**

A single L293D chip can independently control two DC motors, reducing the number of components needed in small multi-motor projects.

- **Expandability and Flexibility**

The system can be expanded by adding more motor drivers or using a motor driver shield. It's compatible with various sensors, camera modules, and communication devices like Wi-Fi, Bluetooth, and GPS.

- **Ideal for IoT and AI Projects**

With internet connectivity and the ability to run AI models, this setup is great for developing smart robots and IoT-based automation.

6.2 APPLICATION

1. Agricultural Robots

These robots can navigate fields to **detect diseases**, **spray fertilizers**, or **monitor soil conditions**. With the Raspberry Pi's processing power and motor control capabilities, these robots can operate autonomously or semi-autonomously.

2. Line-Following and Obstacle-Avoiding Robots

Common in educational and industrial settings, these robots use sensors and camera input processed by the Raspberry Pi to follow paths or avoid obstacles in real time.

3. DIY Robotics Projects

Ideal for **students**, **hobbyists**, and **makers**, this setup is often used to build smart cars, robotic arms, rovers, and automation projects for learning or competitions.

4. Home Automation Systems

The motor driver can control devices like **automated curtains**, **door locks**, **window openers**, and more, bringing physical actuation into home automation.

5. Smart Surveillance Rovers

Mobile robots equipped with cameras can patrol indoor or outdoor areas, stream video, and detect motion or intruders. The Raspberry Pi handles video streaming and decision-making.

6. Smart Wheelchairs or Assistive Mobility Devices

Controlled via joystick, voice commands, or even eye tracking, the Raspberry Pi and motor driver system can power motorized wheelchairs for enhanced accessibility and safety.

7. Prototyping Autonomous Vehicles

Small-scale models of self-driving cars can be built using this system, incorporating cameras, sensors, and machine learning for navigation and control.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The Smart Spraying Rover for Crops project successfully demonstrates how automation, machine learning, and intelligent control systems can significantly improve modern agricultural practices. By integrating a camera-based disease detection system with a machine learning model and an autonomous navigation platform, the rover is able to identify plant diseases in real-time and perform precise pesticide spraying only where necessary. This selective spraying mechanism helps minimize chemical usage, reduce environmental impact, and lower overall farming costs.

The system design—featuring a Raspberry Pi/microcontroller, motor driver, pesticide pump, and image-processing framework—provides a reliable and efficient solution for field operations. The simulation results show that the disease detection model can classify healthy and diseased leaves effectively, validating the core concept of the project. The rover's architecture ensures scalability, adaptability, and future enhancement, such as IoT integration or advanced farm data analytics.

Overall, this project offers a practical and innovative step toward smart farming, addressing major agricultural challenges like labor dependency, excessive pesticide use, and delayed disease detection. The outcomes of this work indicate strong potential for real-world deployment, contributing to higher crop yield, improved sustainability, and enhanced decision-making in agriculture.

7.2 FUTURE SCOPE

The future scope of the Autonomous Smart Spraying Rover for Precision Pesticide Application is very promising due to the rapid development of robotics, artificial intelligence, Internet of Things (IoT), and smart agriculture technologies. The system can be further improved to provide higher efficiency, better accuracy, and fully autonomous agricultural operations.

In future, advanced Artificial Intelligence (AI) and Deep Learning algorithms can be integrated into the rover for more accurate disease detection and crop analysis. Improved machine learning models will help the system identify multiple crop diseases, nutrient deficiencies, pest attacks, and weed growth with higher precision and faster response time.

The integration of GPS and autonomous navigation systems can make the rover completely self-operating in large agricultural fields. GPS-based path planning and automatic route optimization will help the rover cover the entire field efficiently without human intervention. This will reduce operation time and improve spraying accuracy.

Future versions of the rover can also include IoT-based monitoring systems for real-time data transmission and remote operation. Farmers will be able to monitor crop conditions, spraying activity, battery status, and rover location through mobile applications or cloud platforms. This will support smart farming and digital agriculture practices.

The system can be upgraded with advanced sensors such as:

Soil moisture sensors

Temperature and humidity sensors

Gas sensors

Multispectral and thermal cameras

These sensors will help in complete crop health monitoring and environmental analysis.

Solar power technology can also be integrated into the rover to improve energy efficiency and reduce battery dependency. Solar charging systems will allow longer operation time in agricultural fields and support eco-friendly farming practices.

In future, swarm robotics technology may be used where multiple autonomous rovers work together in large farms. This can significantly increase spraying speed, reduce labor requirements, and improve agricultural productivity.

The spraying mechanism can also be improved using variable-rate technology (VRT), where the amount of pesticide sprayed is automatically adjusted according to disease severity and crop conditions. This will further reduce chemical wastage and environmental pollution.

The rover may also be integrated with cloud computing and data analytics systems to maintain field records, analyze crop health trends, and support predictive farming decisions. Farmers can use collected data for better crop management and yield improvement.

Thus, the Autonomous Smart Spraying Rover has great future potential in modern agriculture. With advancements in automation, AI, robotics, and IoT technologies, the system can become a fully intelligent farming solution that improves crop productivity, reduces farming costs, minimizes chemical usage, and supports sustainable agriculture.

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